

Introduction to Artificial Intelligence

Week 7

Agenda

- Monte Carlo Tree Search (Background)
- Connect Four Game
- MCTS (Algorithm)
- Code for MCTS connect four agent

MCTS (Background)

- In games like Chess, Starcraft II, and many more, a heuristic method called Monte Carlo Tree Search is employed to choose the **best course of action** given the **current environmental conditions**.

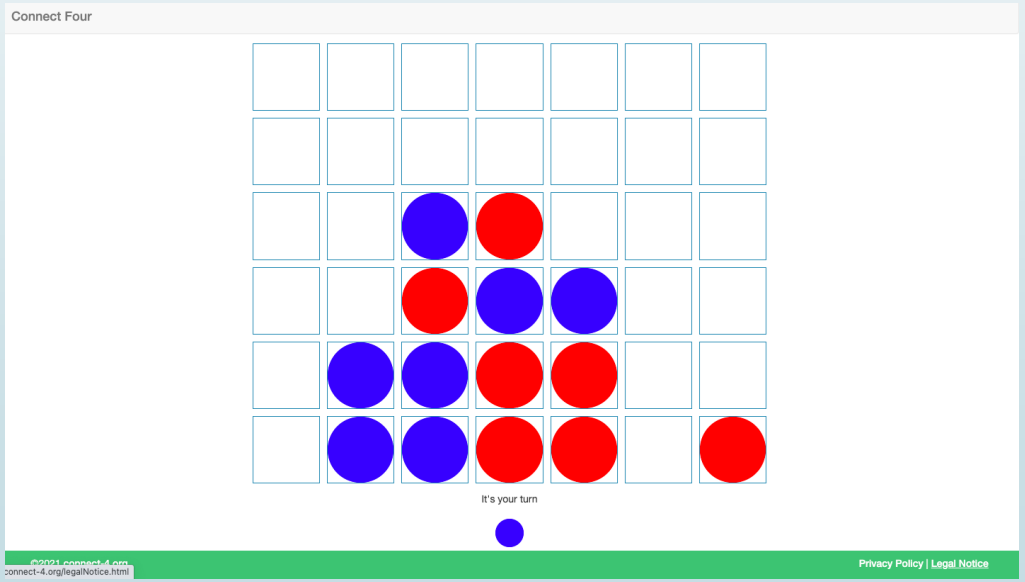
- Monte Carlo Tree Search (MCTS) **doesn't** require an **evaluation function** for states; it **learns** from **general gameplay patterns**. It's used in games with sequences of moves, where **each depth** represents a **new move**. Initially designed for two-player games, it's also applied to one-player games.

Connect 4?

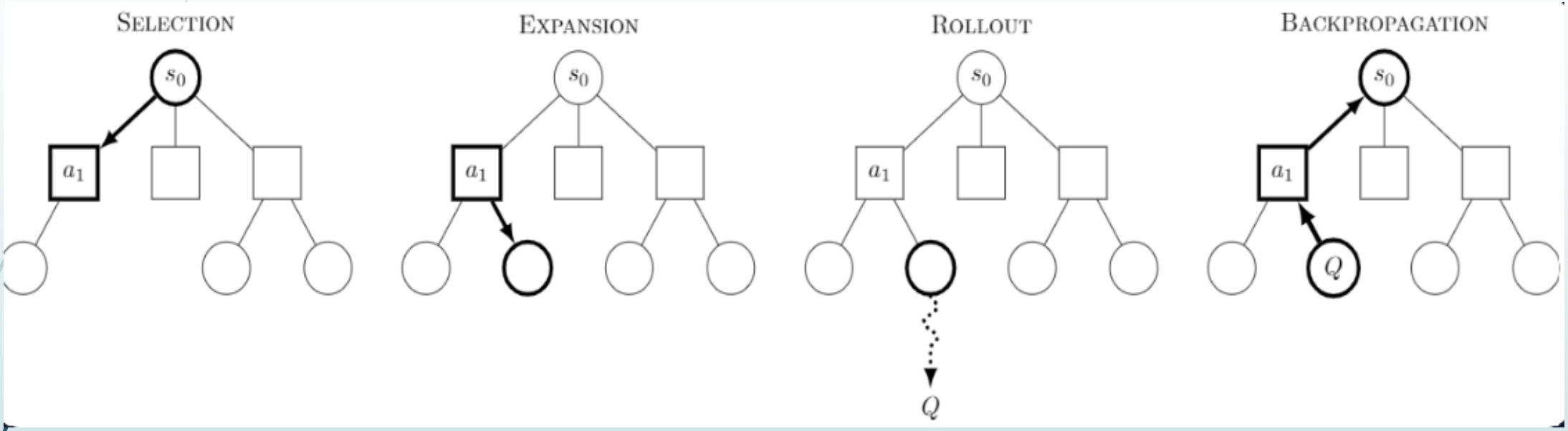
Connect 4 is a **6x7 board game** where players drop their colored pieces to form a sequence of **4 horizontally**, **vertically**, **or diagonally**. The first player to achieve this wins. The pieces should be dropped only from the bottom to up (not at random row).

YouTube

Play Game



MCTS Algorithm



Code

Follow the link below, copy the colab notebook, and complete the missing lines of code.

MCTS lab